

Unit 106 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 88.6 ac, BAF 20 variable-radius cruise on a 300 ft grid, 41 plots, 437 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	213 sq ft/ac (SE 3.1 %)	Type	Sierran mixed conifer
Trees per acre	95	CWHR size / density	4 / D (cover 85 %)
QMD	20.3 in	SDI	272 (47 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	207 sq ft/ac 22.9 in	BA QMD	6 sq ft/ac 7.0 in
TPA	73	TPA	22
Volume	47.1 MBF/ac (8568 cu ft)	Volume	2.4 green tons/ac (80 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~159 sq ft/ac BA	Sampling error	6.3 % at 95 % (t = 2.02, n = 41)
Species (BA)	WF 68, PP 65, IC 32, DF 29, SP 19	Stratum standard	40 % - MEETS; 41 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	9	18.8	4.4	56	0.3
8-11.9	12	10.2	5.9	112	0.6
12-15.9	39	18.5	19.0	512	2.8
16-19.9	55	15.2	26.8	860	4.7
20-23.9	49	9.3	23.9	837	4.6
24-27.9	64	8.6	31.2	1250	6.9
28-31.9	53	5.3	25.9	1133	6.2
32-35.9	51	4.0	24.9	1125	6.2
36-39.9	28	1.8	13.7	687	3.8
40-43.9	29	1.5	14.1	710	3.9
44-47.9	11	0.5	5.4	302	1.7
48-51.9	15	0.5	7.3	428	2.4
52-55.9	6	0.2	2.9	163	0.9
56-59.9	5	0.1	2.4	138	0.8
60-63.9	11	0.3	5.4	335	1.8
All	437	94.7	213.2	8648	47.6

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	139	29.2	67.8	20.6	2777	15.3
PP	134	27.8	65.4	20.7	2700	14.8
IC	66	11.9	32.2	22.2	1314	7.2
DF	59	13.9	28.8	19.5	1134	6.2
SP	39	11.8	19.0	17.2	723	4.0
All	437	94.7	213.2	20.3	8648	47.6

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 106: plots on the 300 ft grid (orange = QA flag)

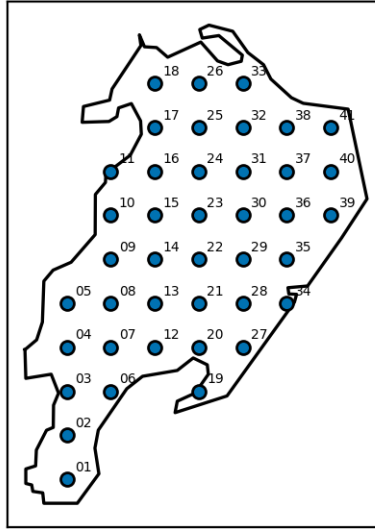


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
106-01	9	180	133	ok
106-02	7	140	34	ok
106-03	12	240	77	ok
106-04	13	260	32	ok
106-05	13	260	31	ok
106-06	11	220	44	ok
106-07	12	240	82	ok
106-08	8	160	17	ok
106-09	10	200	26	ok
106-10	14	280	17	ok
106-11	13	260	32	ok
106-12	8	160	177	ok
106-13	7	140	26	ok
106-14	10	200	9	ok
106-15	13	260	39	ok
106-16	11	220	34	ok
106-17	9	180	142	ok
106-18	11	220	64	ok
106-19	12	240	79	ok
106-20	12	240	90	ok
106-21	7	140	19	ok
106-22	12	240	67	ok
106-23	12	240	15	ok
106-24	11	220	26	ok
106-25	10	200	32	ok
106-26	13	260	81	ok
106-27	11	220	100	ok
106-28	10	200	96	ok
106-29	10	200	21	ok
106-30	10	200	62	ok
106-31	11	220	53	ok
106-32	11	220	10	ok
106-33	12	240	36	ok
106-34	11	220	96	ok
106-35	12	240	15	ok
106-36	12	240	13	ok
106-37	11	220	37	ok
106-38	11	220	36	ok
106-39	3	60	14	ok
106-40	10	200	27	ok
106-41	12	240	13	ok